

Exercise 1: Setting Up JUnit

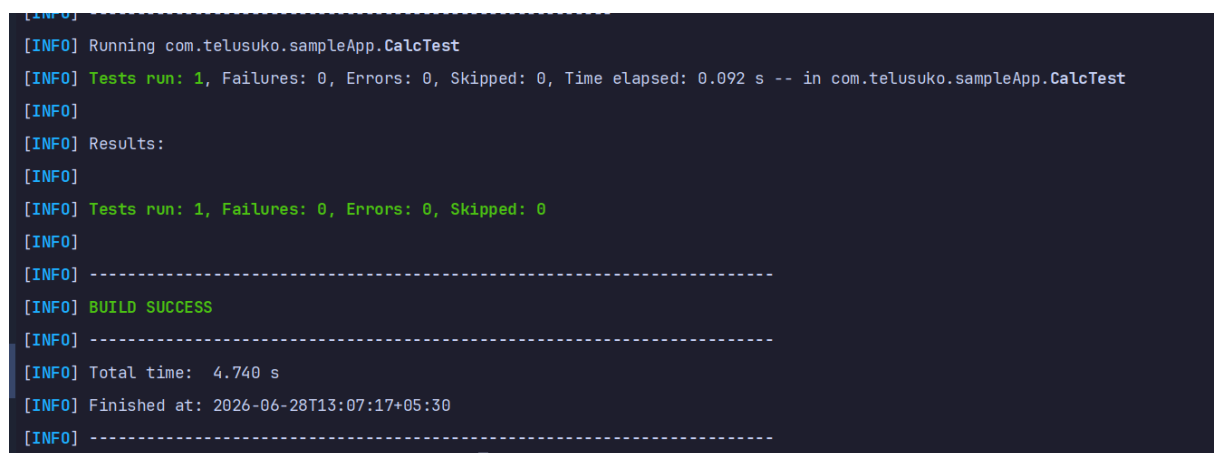
```
✓ pom.xml (MavenProject) ×
11 <properties>
12     <maven.compiler.source>21</maven.compiler.source>
13     <maven.compiler.target>21</maven.compiler.target>
14     <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
15 </properties>
16 <dependencies>
17     <!-- Source: https://mvnrepository.com/artifact/org.junit.jupiter/junit-jupiter-api -->
18     <dependency>
19         <groupId>org.junit.jupiter</groupId>
20         <artifactId>junit-jupiter-api</artifactId>
21         <version>5.14.3</version>
22         <scope>test</scope>
23     </dependency>
24     <dependency>
25         <groupId>junit</groupId>
26         <artifactId>junit</artifactId>
27         <version>RELEASE</version>
28         <scope>test</scope>
29     </dependency>
30 </dependencies>
31 </project>
```

```
✓ pom.xml (MavenProject)  CalcTest.java ×
1 package com.telusuko.sampleApp;
2 import org.junit.jupiter.api.Test;
3
4 import static org.junit.Assert.assertEquals;
5
6 ▶▶ public class CalcTest {
7     @Test // we use this annotation when we are testing. and we need to add required file in pom.
8     ▶ void testCalc(){
9         Calc calc = new Calc();
10         int actualResult = calc.divide(10,5);
11         int expectedResult = 2;
12         assertEquals(expectedResult,actualResult); // it means comparision btw the expectation an
13     }
14 }
15
```



```
1 package com.telusuko.sampleApp;
2
3 public class Calc {
4     > int divide(int n1, int n2) { return n1/n2; }
5 }
6
7
8
```

Output:



```
[INFO] Running com.telusuko.sampleApp.CalcTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.092 s -- in com.telusuko.sampleApp.CalcTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.740 s
[INFO] Finished at: 2026-06-28T13:07:17+05:30
[INFO] -----
```

Exercise 3: Assertions in JUnit

```
1 package com.telusuko.sampleApp.Exercise3;
2 import org.junit.jupiter.api.Test;
3
4 import static org.junit.jupiter.api.Assertions.*;
5 ✓ public class AssertionsTest {
6     @Test
7     ✓ void testAssertions(){
8         assertEquals( expected: 5, actual: 2+3);
9         assertTrue( condition: 5>3);
10        assertFalse( condition: 5<3);
11        String name = null;
12        assertNull(name);
13        Object obj = new Object();
14        assertNotNull(obj);
15
16    }
17
18 }
19
```

output:

```
[INFO] T E S T S
[INFO] -----
[INFO] Running com.telusuko.sampleApp.Exercise3.AssertionsTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.090 s -- in com.telusuko.sampleApp.Exercise3.AssertionsTest
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.433 s
[INFO] Finished at: 2026-06-28T14:25:22+05:30
[INFO] -----
PS D:\DESKTOP\Java Maven Practice\MavenProject>
```

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

```
package com.telusuko.sampleApp.Exercise4;
```

2 usages

```
public class Calculator {
```

2 usages

```
     int add(int a,int b){
```

```
        return a+b;
```

```
    }
```

```
}
```

```
package com.telusuko.sampleApp.Exercise4;
import org.junit.jupiter.api.*;
import static org.junit.jupiter.api.Assertions.*;
public class CalculatorTest {
    Calculator calculator;
    @BeforeEach
    void setUp(){
        calculator = new Calculator();
        System.out.println("Setup executed");
    }
    @Test
    void testAddition(){
        int a = 10;
        int b = 20;
        int result = calculator.add(a,b);
        assertEquals(30,result);
    }
    @Test
    void testAddition2(){
        int result = calculator.add(50,30);
        assertEquals(80,result);
    }
    @AfterEach
    void tearDown(){
        System.out.println("Teardown executed");
    }
}
```

Output:

```
[INFO] Running com.telusuko.sampleApp.Exercise4.CalculatorTest
Setup executed
Teardown executed
Setup executed
Teardown executed
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.109 s -- in com.telusuko.sampleApp.Exercise4.CalculatorTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.464 s
[INFO] Finished at: 2026-06-28T14:52:15+05:30
```